Mining Visual Signatures of AI-Generated Images vs Real Images

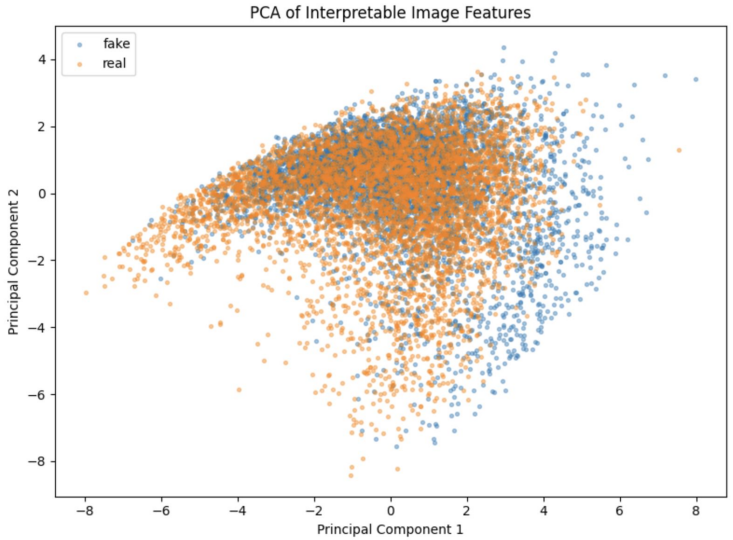
Motivation

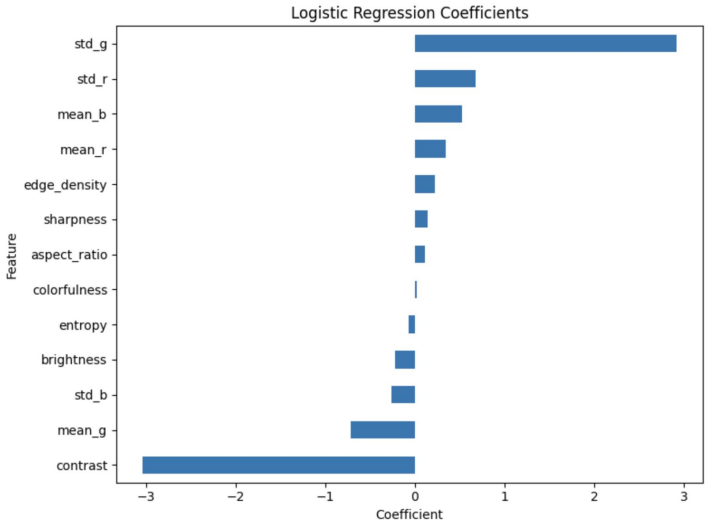
- AI-generated images are becoming harder to identify visually.
- Can simple interpretable image features distinguish fake from real images?
- Focus: transparent data mining instead of deep learning.

Data & Design

- **60,000 images**: 30,000 real, 30,000 AI-generated
- Balanced **train/test** split
- Features: brightness, contrast, entropy, sharpness, edge density, colorfulness, RGB statistics

Panel A: PCA scatter





Panel B: Logistic regression coefficients

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Findings, Tradeoff, and Takeaway

Main Findings

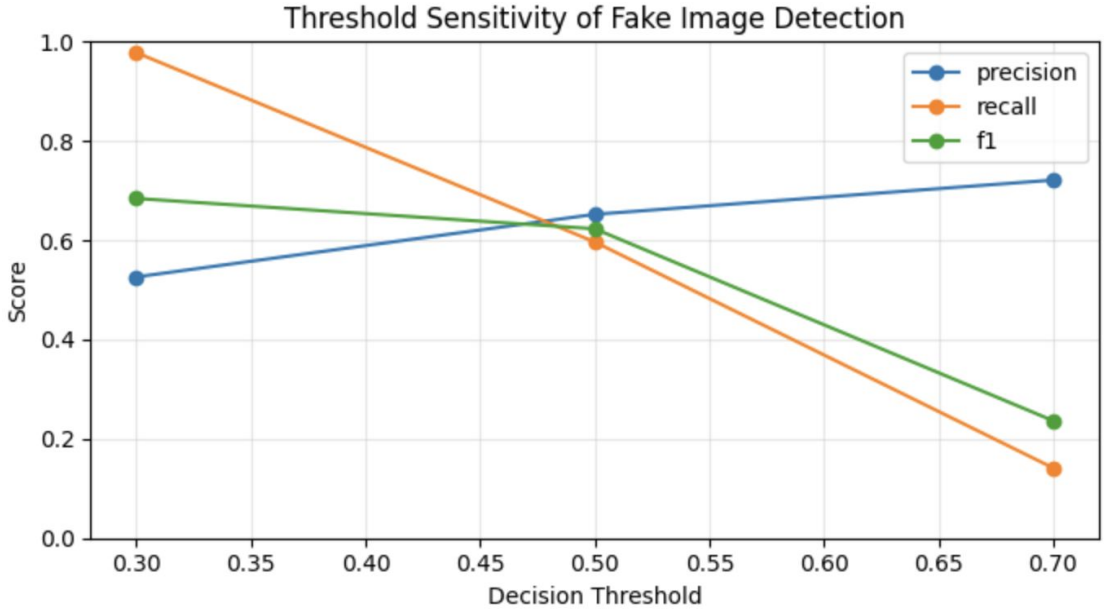
- PCA shows **substantial overlap**, but real and fake images shift in opposite directions.
- Fake images tend to be higher in **sharpness**, **colorfulness**, **channel variability**, and **edge density**.
- Logistic regression achieved **0.639 accuracy** and **0.693 ROC-AUC**.

Threshold Tradeoff

- Lower threshold: catches most fake images, but with more false alarms
- Higher threshold: improves precision, but misses many fake images

Takeaway

- Handcrafted features reveal meaningful visual signatures
- But they are not enough for fully reliable detection



Threshold Sensitivity of Fake Image Detection

Repo Link: https://github.com/Janakrish1/Data_Mining_Project

Dataset: ai-generated-images-vs-real-images